Module Title: Machine Learning

Understanding Support Vector Machines (SVM):

A Visual and Practical Tutorial on Linear and Non-linear Classification

Student Name: **Bhargav Chundu**

Student ID: **24150503**

GitHub Link: https://github.com/BhargavChundu/Individual-Machine-Learning-.git

Introduction

Machine learning is one of the essential and powerful tools in data analysis that makes systems learn patterns from the information shared, predict as per information, and make decisions without any prior/hardcoded programming. Classification algorithms are one of the essential machine learning techniques used in tasks where the goal is to assign data points to predefined categories. Support Vector Machine (SVM) is one of the most used and theoretically sound classification methods.

Support Vector Machines are supervised learning algorithms which seek to find the optimal decision boundary (or hyperplane), that separates points belonging to different classes. SVM is unlike most of the classification algorithms in that SVM aims to separate classes and also maximize the margin between those classes. This margin is the separation distance between the hyperplane decision boundary and the closest data points in either class called support vectors. SVM achieves improved generalisation and robustness to unseen data by maximizing this margin.

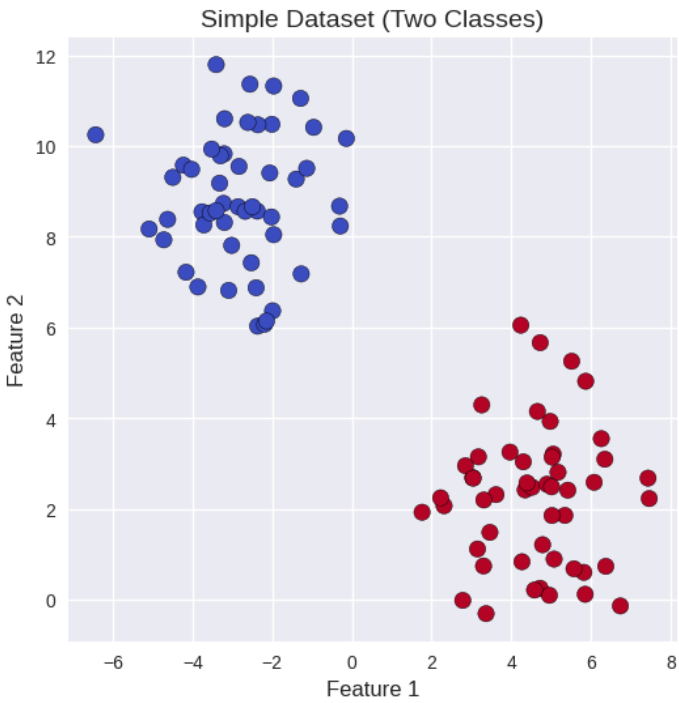
One of the strengths of SVM is that it performs well on both linearly separable and non-linearly separable datasets. A linear SVM creates a straight line or hyperplane as the decision boundary, however many real world datasets are complex and cannot be separated using a straight line. The challenge with SVM is that it cannot help for data that are not linearly separable. Popular choices of k where he function are linear, polynomial and radial basis function (RBF) kernels, each offering different amounts of flexibility.

So, this tutorial will cover the theory of SVMs as well as code demonstrating how to implement an SVM from scratch. We show that SVM leads to different results under multiple scenarios, including linearly separable data and non-linear synthetic datasets. Additionally, we can compare the effect of various kernel functions and hyperparameters on how well the model performs with visualisations and evaluation metrics. We will

also compare SVM to other types of classification algorithms, which we hope will allow the reader to understand some key practical strengths and weaknesses associated with SVM while at the same time make it an easy method to grasp.

2. Dataset and Problem Description

Figure 1: Linearly separable dataset with two distinct classes.



Two synthetic datasets were created to explore the behaviour and performance of Support Vector Machines (SVM): a linearly separable dataset and a non-linearly separable dataset. These datasets offer a controlled environment to study SVM model performance in relation to brand data distributions and complexities.

2.1 Linearly Separable Dataset

The initial dataset contains two well defined clusters, generated using a Gaussian distribution. Clusters are classes here labelled 0 and 1 respectively Figure 1: Here, the two classes are easily separable in the feature space; we can classify them with a straight line.

The two features (that's what they're called) Feature 1 and Feature 2 correspond to two numerical inputs that indicate the position of each datapoint in a two-dimensional space. This dataset is nice to use because the two classes are very clearly separated, which makes a simple linear SVM good at classifying this data, as finding a liner decision boundary that separates the two will effectively classify each sample correctly.

This dataset provides a baseline scenario to see what SVM does when we give it a simple linearly separable classification problem.

2.2 Non-linearly Separable Dataset

The second dataset is produced from a “moons” distribution, which generates two interleaved crescent-shaped clusters. Unlike the previous dataset, the classes here are not linearly separable. As shown in Figure 2, the data points overlap so that no linear separator exists for both classes.

This dataset adds a new level of complexity and is good for Displaying the limitations of linear models. It emphasizes the requirement for a more flexible decision boundary which can better represent the nonlinear structure of the dataset.

Having both datasets allows for a comparative study of how SVM behaves in these two cases. The first dataset is linearly separable requiring only a simple linear boundary to represent the target function, while the second one (the non-linear dataset) shows that for this case we need kernel methods to transform our inputs into higher-dimensional spaces so as to represent its more complex behaviour.

3. Theory of Support Vector Machines

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. The main goal of SVM is to find an optimal decision boundary, termed as hyperplane in the context of SVM, that separates data points from different classes. Unlike many traditional classifiers, SVM attempts not only to separate the data but also maximise the margin between classes which aids in improving generalisation on unseen data.

3.1 Decision Boundary and Hyperplane

A hyperplane is just a straight line that divides the classes in a two-dimensional feature space. In higher dimensions the hyperplane can be a plane or even an hyper-surface. The objective of SVM is to identify the hyperplane that provides optimal separation between the data, such that points belonging to different classes reside in opposing regions.

But we know there exists multiple possible hyperplanes to separate this data. It chooses the best one by a specific criteria, max margin.

3.2 Margin Maximisation

The margin is the space between the closest points of either class from the decision boundary. These nearest points to the decision boundary are crucial as they affect where the actual decision boundary lies.

SVM tries to maximize this margin, i.e., selects hyperplane which is as far away from nearest data points in both classes. This, in turn means that the model becomes less sensitive to small variations/noise in the data as a result of which larger margin generally results in better generalisation.

By maximising the margin, classification is transformed into an optimisation problem, where the algorithm strives to trade-off between separation and robustness.

3.3 Support Vectors

The support vectors are the data points that lay closest to the decision boundary. Because they require those points to determine the orientation and arrangement of hyperplane. While other data points would have minor impact on the boundary, support vectors directly influence the solution of the SVM model.

A key feature of SVM is that it only relies on these support vectors and not the whole dataset. By constructing a decision boundary with only examples from the subset of data makes this model fast and robust.

3.4 Linear vs Non-linear Classification

For linearly separable data, a straight hyperplane correctly classifies the data. But in practice this is rarely achievable, as most real-world datasets are not linearly separable, so a table does an acceptable job of separating the dataset[1].

In order to solve this limitation, SVM adopted the idea of non linear in class with kernel functions. SVM does this by transforming the data in a way that allows it to find a linear boundary in what is called a higher-dimensional space.

3.5 Kernel Trick

Among several innovations in SVM, the kernel trick is one of the most important which enables it to fit a non-linear dataset without using multiple features. The kernel function calculates the relationship between pairs of data points as if these data points were transformed into some higher dimensional space, without actually making that explicit transformation.

Common kernel functions include:

- Kernel functionl: It includes the most common kernels such as linear kernel which is used if you have a linearly separable data

- Polynomial kernel : Enables curved decision boundaries
- Radial Basis Function (RBF) kernel: Offers very flexible, nonlinear boundaries.

Out of these, the RBF kernel is commonly used since it can model complex patterns while generalizing well.

3.6 Generalisation and Robustness

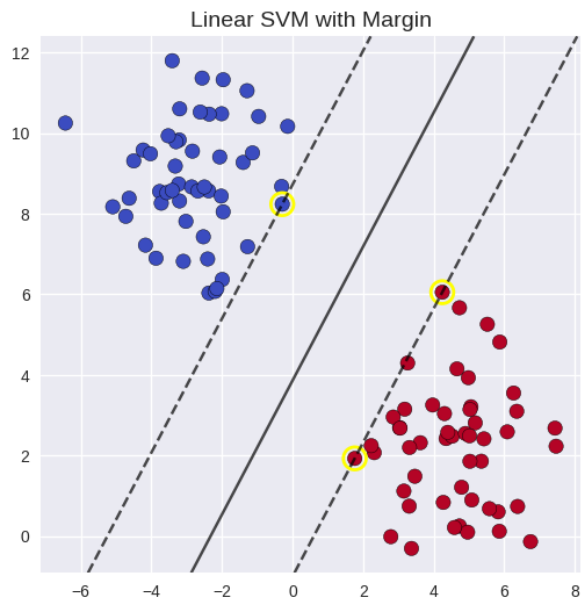
SVM is well-known for its theoretical underpinning of maximum margin, which yields better generalisation performance. Due to only relying on the most relevant data points (support vectors) and a large margin between different classifications, SVM is less prone to overfitting like many other models.

That said, the SVM performance is heavily reliant on the kernel and hyperparameters. Choosing the right values is key to getting good performance, especially in case of non-linear classification problem.

4. Linear Support Vector Machine

To illustrate the basic functionality of Support Vector Machines, a linear sv machine model was fitted on the synthetic linearly separable dataset described in Section 2. As can be seen, our model is able to detect a straight decision boundary separating the two classes with high accuracy.

Figure 2: Linear SVM showing decision boundary, margin, and support vectors.



As shown in Figure 2, the solid line is the decision boundary that separates the regions in feature space corresponding to each of two classes. The boundary is represented by the solid line with dashed lines on either side indicating the margin, which comes from each class next to the nearest data points. The goal of the SVM is to increase this margin, so these points are at a maximum distance from the boundary.

Support vectors: The actual data points on the margin of SVM are highlighted and referred to as support vectors. These points are important for defining the model, since they determine where the decision boundary lies. Points like the other data points, which are too far away from that boundary, do not affect the final model.

This is mainly because the dataset is very linear, so a linear SVM would have been seen as a perfect classifier. The data here is clearly linear separable, so the model is able to achieve a nice decision boundary without any misclassification. This shows the power of linear SVM for simple classification problems.

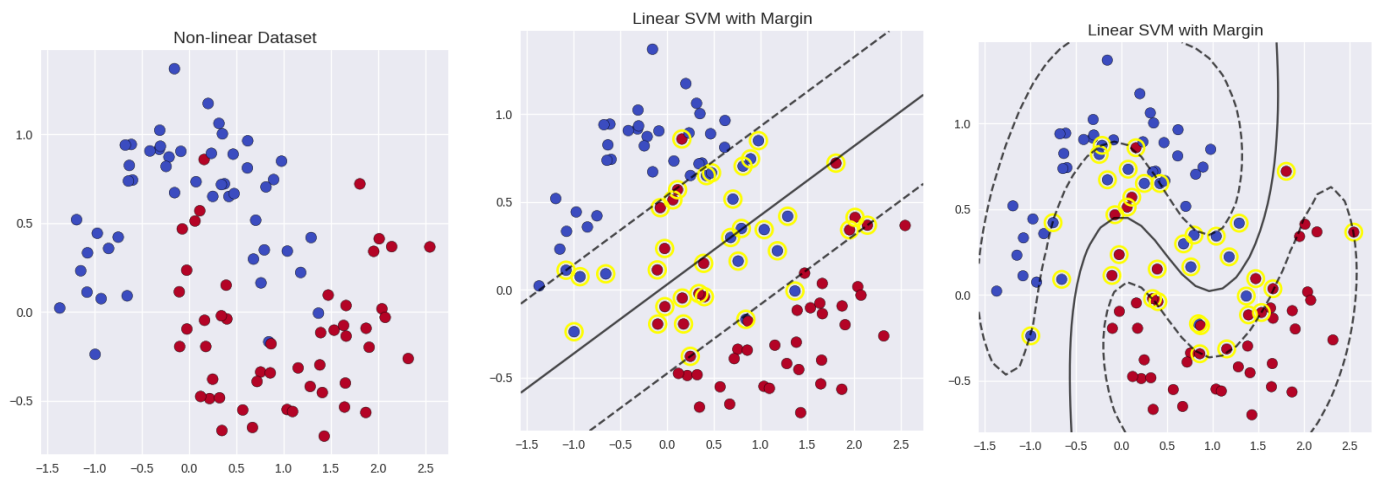
In addition, the margin maximisation principle makes sure that the model is accurate that is robust. The classifier is less sensitive to small variations in the data, allowing for a better generalisation of unseen samples by keeping as large of a margin as possible.

However, linear SVM has drawback when it comes to the more complex distributions of data as shown in above example. Read on for the next section where we can see how linear SVM fails in case of a non linearly separable data and why we need such advanced techniques.

5. Limitations of Linear SVM

The linear SVM works quite well for linearly separable data, however, is not useful with complex datasets. To demonstrate this, the linear SVM was run on a non-linearly separable dataset— "moons"—which we discussed earlier.

Figures 3,4,5; Non-linearly separable dataset , Linear SVM failing to separate non-linear data.



The dataset, illustrated in Figures, includes two interlocking classes of crescent shapes that cannot be separated with a linear boundary. But that structure is a huge problem for linear classifiers; no line separating the two classes can be completely free of error.

Now, if you train a linear SVM on this dataset, the resulting classifier will not be able to capture the embedded pattern in data. For example the five-class separation as shown in Figure 5 can be done using a straight line

but this will lead to a significant amount of misclassification. It is particularly evident in the first example, where the boundary cuts through the curved structure of the dataset — indicating that the model couldn't find a way to adapt to non-linear relationships within its training.

This is a limitation, as linear SVM requires the data to be linearly separable in its original input context. In cases where this is not true, the model becomes too limited to capture the data distribution accurately.

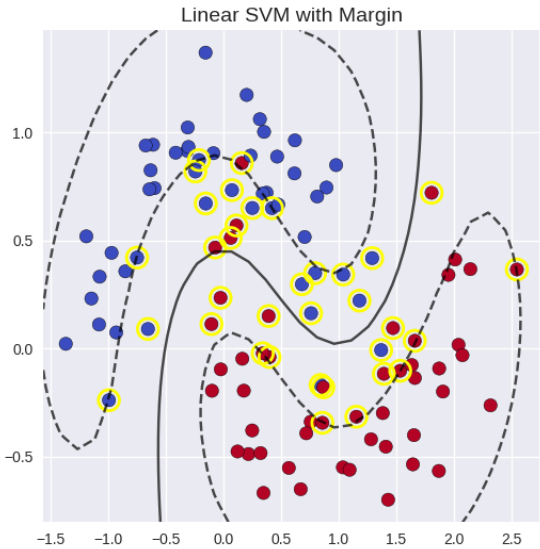
In such cases, the linear SVM will fail, indicating a necessity of using methods that can classify non-linear patterns. In practice, data is usually too complex and hardly follows a simple linear form. Hence, using only linear models will show worse results.

Support Vector Machines address this limitation leveraged by introducing the concept of kernel functions (K), which allow for non-linear decision boundaries to be constructed. In the next part we discuss how SVM can use kernels to transform feature space such as it is able to successfully classify non-linear data.

6. Kernel Trick and Non-linear SVM

The kernel trick, a powerful approach in the Support Vector Machine arsenal, is introduced to mitigate the challenges of linear SVM on non-linearly separable datasets. This method enables the model to learn non-linear decision boundaries without moving data into a higher-dimensional space.

Figure 6: RBF kernel producing a non-linear decision boundary.



With kernel trick, the idea is to map the original input features into a higher dimensional feature space that makes the data linearly separable. Rather than applying this transformation directly (which can be

computationally expensive), SVMs use a kernel function to calculate the relationships between data points as if they already lived in that higher-dimensional space.

The Radial Basis Function (RBF) kernel is one of the most commonly used kernel functions, owing to its flexibility and performance. The RBF kernel can represent complex decision boundaries due to its ability to create very smooth and well fitted boundaries fitting even more the highly nonlinear data.

For instance, as shown in Figure 6, utilizing the RBF kernel on the non-linear moons dataset allows the SVM model to create a curved decision boundary that closely follows the data structure. Compared to the linear model which does not properly separate the classes, RBF is very successful distinguishing classes with a low error rate.

It is better because the kernel function allows you to learn non-linear relationships of all sorts between data points. You have been trained on data until October of 2023

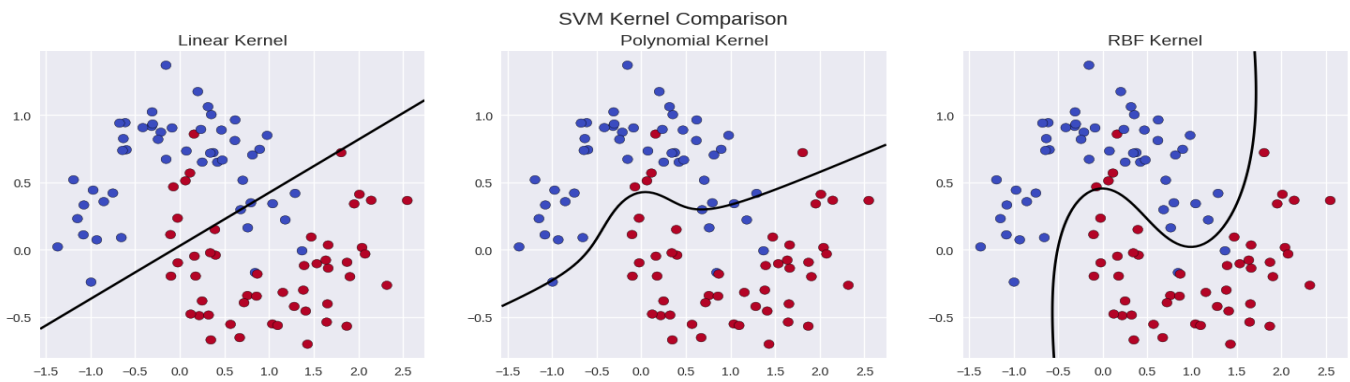
The use of the kernel trick is a significant benefit of SVM, since it provides versatility ranging from simple linear classifications to more complicated non-linear challenges. They balance frequently, though the kernel and its hyperparameters are specific to model efficiency.

Overall, the use of the RBF kernel illustrates how SVM can address the rigidity constraints of linear classification by incorporating non-linear flexibility into its dichotomous boundary. This is what gives SVM its strength and versatility for many real-world machine learning problems.

7. Kernel Comparison

To explore the effect of diverse kernel functions on model accuracy, an experiment was performed with three conventional kernels: linear, polynomial and RBF. By using the same data (non-linearly separable), a direct visual and quantitative comparison was made for all three models.

Figure 7: RBF kernel producing a non-linear decision boundary.



As shown in Figure 7, linear kernel results in a straight decision boundary that does not fit the curved structure of this dataset. This leads to a number of incorrectly classified points, and we can say that linear models are not appropriate for complex non-linear patterns.

The polynomial kernel provides a curved space for the decision function in contrast to the linear kernel, allowing it to fit better with data. But for any chosen degree the resulting boundary could still be inflexible or unnecessarily complex. The polynomial kernel in this case improves the separability but does not fully reflect the underlying distribution of points.

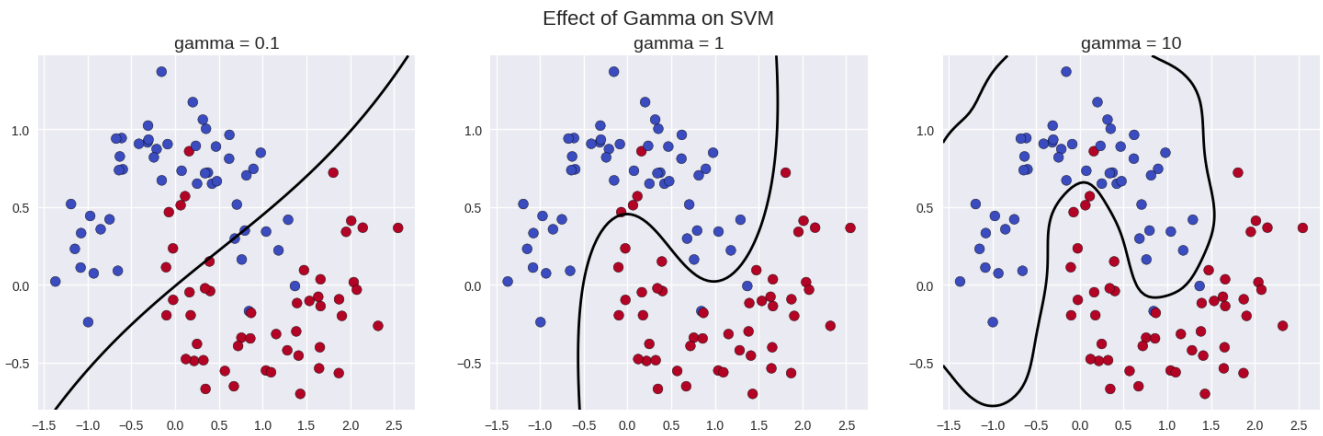
The RBF kernel, on the other hand, creates a continuous and adaptable non-linear decision boundary that conforms to the contour of the data. The RBF model shows the most accurate separation, with a minimum of misclassifications as can be seen on the graph. This shows its potential to manage complex patterns extremely well.

As the output of an SVM model heavily relies on the chosen kernel, these observations further emphasize the importance of selecting an appropriate algorithm when training a machine learning classifier. Simplistic kernels like the linear kernel may be computationally cheaper, but can be simplistic when it comes to complex data. In contrast, flexible kernels such as RBF can fit complex relationships, but they involve hyperparameter tuning.

In general, the comparision validate that RBF kernel provides the best trade off between flexibility and generalisation for the dataset used in this study.

8. Hyperparameter Analysis (Effect of Gamma)

Figure 7: Effect of gamma on decision boundary complexity.



Besides a good choice of kernel, the performance of Support Vector Machine depends largely on its hyperparameters. The two main types of kernel functions that are popular in SVMs are linear and radial basis function (RBF) based, The most important hyper parameter for RBF kernel is gamma.

To study the impact of this parameter, we trained the RBF SVM using multiple gamma values. Results are shown in the following figure, which shows how the shape of decision boundary changes for various value of gamma.

For small values of gamma (e.g. 0.1), the model creates a very smooth and simplistic decision boundary. Here, each point has a large region it affects, leading to the boundary that may not encapsulate the nature of the data. This may result in underfitting because the model would not learn significant patterns in the data.

A moderate value of gamma (e.g. 1) causes the decision boundary to adapt to the dataset structure in a way that better reflects reality. Such an approach, usually leads to a better classification performance, due to code model finding its balance between dedicating simplicity and accuracy. So generalisation does happen here and we can see that overfitting is not happening.

With a large gamma (e.g. 10), the boundary/radius becomes very jagged and tightly binds around all the training data. The model pays too much attention to an individual sample, since the area of influence for each data point is very small. This leads to overfitting, or when the model picks up more noise than the underlying pattern in the data.

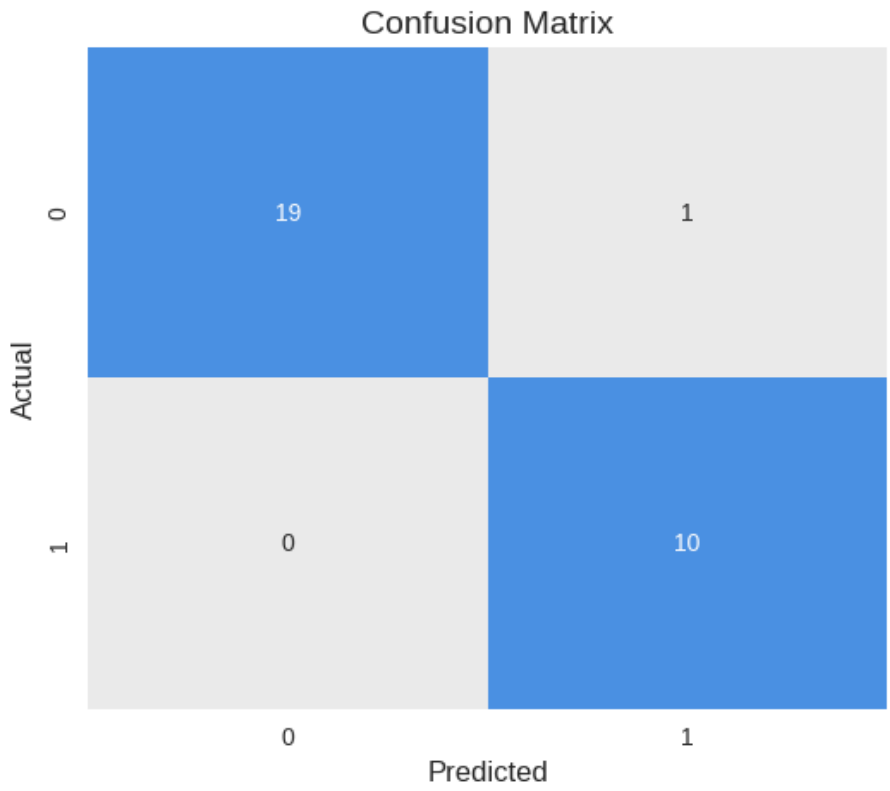
Hyperparameter tuning is an important part of SVM as evidenced by this experiment. Answering the questing, what is optimal performance for a given gamma would be trained on data until October of 2023 as it determines a balance between underfitting and overfitting.

9. Model Evaluation

In addition to qualitative inspection, quantitative metrics were employed to assess the discriminative ability of the different SVM models. The dataset should be split into training and testing sets, and the model performance could be evaluated via classification accuracy and confusion matrix.

RBF kernel achieve highest accuracy of 0.97, is much greater than linear and polynomial kernels as shown in results Linear Kernel: 0.87 (It was able to capture a bit of structure in the data but failed to model dearly non-linear patterns.) The polynomial kernel returned an accuracy of 0.83 which means that it did not perform well in this particular case, possibly due to parameter being too high or the kernel's flexibility being insufficient.

Figure 8: Confusion matrix showing classification performance.

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Linear Accuracy: 0.87
Polynomial Accuracy: 0.83
RBF Accuracy: 0.97
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Figure 8 references the confusion matrix that gives a more detailed perspective of model performance. It depicts the correct and incorrect predictions made by the classifier. Correct classifications are shown along the diagonal and misclassifications off the diagonal.

Looking at the confusion matrix, it can be seen that most predictions are on its diagonal, which means we predict our dependent correctly. And there are only few misclassifications in the confusion matrix, that you can see confirms that model is working as expected. Apart from this, using a two-colour visualisation makes it easier to differentiate between the correct predictions and errors making our interpretation very easy!

These findings align with the previous visualisations. In contrast, the RBF kernel molded the decision boundary to better fit the data structure, providing a more accurate classification. Using non-parameteric models (polynomial and linear) didn't seem to fit the underlying structure of the dataset, but it could be used in situations where linear relationships were apparent.

In summary, this analysis shows that the decision of which kernel and hyperparameters to use heavily influences model performance. In summary the RBF kernel is powerful when tuned correctly, being flexible enough to model complex data while generalising extremely well.

10. Conclusion

Support Vector Machines (SVM): Implementation and Theory, In this tutorial, we learned about what Support Vector Machines (SVM) are in terms of their theory as well as application. The study started with a case of linearly separable data where all points were correctly classified by a linear SVM that found the decision boundary with maximum margin. This showed the basic idea of SVM and its power in simple classification tasks.

But, when applied to non-linear separable data, the restrictions of linear SVM were clear. The moons dataset was shown to highlight that a linear decision boundary does not capture the pattern well and leads to misclassification. This pointed to a need for more practical flexibility.

The kernel trick was then introduced to overcome this limitation of SVM, allowing it to work with non-linear data by implicitly mapping the training examples into a higher dimensional space. The most effective kernel among those tested was the Radial Basis Function (RBF) kernel, as it produced a smooth and flexible decision boundary that fit well with the structure of the dataset.

A more complex analysis showed that kernel selection and the tuning of hyperparameters were indeed important. We also found that the comparison of linear vs polynomial vs RBF in different sets performed significantly different based on the chosen kernel. The learned behavior of gamma was also enlightening as it showed how we can control the complexity of our model and provided insight into underfitting vs overfitting.

The results of the evaluation corroborate these findings with respect to accuracy (the RBF kernel managed the highest with 0.97, beating both linear and polynomial). The confusion matrix also confirmed this finding, with a high number of correctly classified observations and very few miss-classifieds.

The above framework may be upgraded to become Support Vector Machines which is a robust and adaptive classification algorithm for simple to highly complicated datasets. They are particularly powerful for non-linear problems through their capability to maximise margin and use of kernel function. However, their performance relies heavily on Christ and opti_sparse proper kernel choice and parameter tuning. SVM gives good generalisation and predictive ability when used properly, so it is one of the useful machine learning algorithms.

References

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